



Striving to be the world's top BESS solution provider of all applications



Energy Storage System Brochure

SYL (NINGBO) BATTERY CO., LTD.

SYL (NINGBO) BATTERY CO., LTD.

Add: No.23 Xingke Zhong Road, Ninghai, Ningbo, China.315609

E-mail: marketing@sylbattery.com

Web: www.sylbattery.com

Tel: +86-400-101-8585



About SYL Battery

SYL is a subsidiary of the Risen Energy Group, focusing on li-ion batteries for more than 10 years, is committed to become the world's leading BESS solutions provider for all applications, with annual production capacity of 15GWh+.

Its products include cell, module, C&I and large-scale BESS. The independently developed liquid-cooled energy storage battery system is the first in China passing UL9540A certification in both the US and China. SYL's projects have been deployed around the world, with rich experience in delivering large-scale BESS projects above 100MW globally, with quality and service widely recognized by worldwide customers.



Utility-Scale BESS Solution-1h

Cabinet Solution



Container Solution



Highlights

- One-stop solution of the utility-scale BESS application
- Various operating modes like VSG, VF and PQ to improve power quality
- Highly integrated design, minimized site installation and CapEx
- 3450kW high power density with modular design Cost and solution benefits in Utility-Scale BESS

Technical Specification

System Model			SU3441E3440MV	
C-rate			1C	
Medium Voltage Power Station	Power of MV Station		3440kVA	
	Transformer	Rated Power	3440kVA	
		Nominal Voltage	6~35kV	
		No.of phases	3	
		Frequency	50/60Hz	
	Rmu	Switch Configuration	Breaker/Load Switch + Fuse Combination	
		Insulating Medium	SF6	
		Voltage Range	6~35kV	
	Auxiliary Transformer		Nominal Power	100kVA
	Converter	Rated Power	1725kW	
		DC Voltage Range	1500Vdc	
		AC Rated Voltage	690Vac	
		Maximum Efficiency	99%	
		Quantity	2	
	Anti-Corrosion		C4	
	IP Rating		IP55	
	Dimensions (W×H×D)		6058×2896×2438mm	
	Weight		18,000Kg	
	Compliance		GB17467, IEC62271-202	
	Communication interfaces		RS485/CAN/Ethernet	
	Communication protocols		Modbus-RTU/Modbus-TCP/IEC61850/IEC104	
	Operating Temperature Range		-25℃~60℃	
	Max. Working Altitude		4000m	
DC Side Option (1)				
Cabinet Option	Nameplate Energy		344.064kWh	
	Dimension (W×H×D)		1300×2428×1300mm	
	Communication interfaces		RS485/CAN	
	Communication protocols		Modbus-RTU	
	Operating Temperature Range		-30~50℃	
	Max. Working Altitude		3000m	
	Weight		3,500Kg	
	Quantity		10	
DC Side Option (2)				
Container Option	Nameplate Energy		3.44MWh	
	Dimension (W×H×D)		6058×2896×2438mm	
	Communication interfaces		RS485/CAN/Ethernet	
	Communication protocols		Modbus-RTU/Modbus-TCP/IEC61850	
	Operating Temperature Range		-30℃~50℃	
	Max. Working Altitude		3000m	
	Weight		34,000Kg	
	Quantity		1	

Utility-Scale BESS Solution-2h

Cabinet Solution



Container Solution



Highlights

- One-stop solution of the utility-scale BESS application
- Various operating modes like VSG, VF and PQ to improve power quality
- Highly integrated design, minimized site installation and CapEx
- 3450kW high power density with modular design Cost and solution benefits in Utility-Scale BESS

Technical Specification

System Model			SU6882E3440MV
C-rate			0.5C
Medium Voltage Power Station	Power of MV Station		3440kVA
	Transformer	Rated Power	3440kVA
		Nominal Voltage	6~35kV
		No.of Phases	3
		Frequency	50/60Hz
	Rmu	Switch Configuration	Breaker/Load Switch + Fuse Combination
		Insulating Medium	SF6
		Voltage Range	6~35kV
	Auxiliary Transformer	Nominal Power	100kVA
	Converter	Rated Power	1725kW
		Maximum DC Voltage	1500Vdc
		AC Rated Voltage	690Vac
		Maximum Efficiency	99%
		Quantity	2
	Anti-Corrosion		C4
	IP Rating		IP55
	Dimensions (W×H×D)		6058×2896×2438mm
	Weight		18,000Kg
Compliance		GB17467,IEC62271-202	
Communication Interfaces		RS485/CAN/Ethernet	
Communication Protocols		Modbus-RTU/Modbus-TCP/IEC61850/IEC104	
Operating Temperature Range		-25℃~60℃	
Max. Working Altitude		4000m	
DC Side Option (1)			
Cabinet Option	Nameplate Energy		344.064kWh
	Dimension (W×H×D)		1300×2428×1300
	Communication interfaces		RS485/CAN
	Communication Protocols		Modbus-RTU
	Operating Temperature Range		-30~50℃
	Max. Working Altitude		3000m
	Weight		3,500Kg
	Quantity		20
DC Side Option (2)			
Container Option	Nameplate Energy		3.44MWh
	Dimension (W×H×D)		6058×2896×2438mm
	Communication interfaces		RS485/CAN/Ethernet
	Communication Protocols		Modbus-RTU/Modbus-TCP/IEC61850/IEC104
	Operating Temperature Range		-30℃~50℃
	Max. Working Altitude		3000m
	Weight		34,000Kg
	Quantity		2

Utility-Scale BESS Solution-4h

Container Solution



Highlights

- One-stop solution of the utility-scale BESS application
- Various operating modes like VSG, VF and PQ to improve power quality
- Highly integrated design, minimized site installation and CapEx
- 3450kW high power density with modular design Cost and solution benefits in Utility-Scale BESS

Technical Specification

System Model			SU13764E3440MV	
C-rate			0.25C	
Medium Voltage Power Station	Power of MV Station		3440kVA	
	Transformer	Rated Power	3440kVA	
		Nominal Voltage	6~35kV	
		No.of phases	3	
		Frequency	50/60Hz	
	Rmu	Switch Configuration	Breaker/Load Switch + Fuse Combination	
		Insulating Medium	SF6	
		Voltage Range	6~35kV	
	Auxiliary Transformer		Nominal Power	150kVA
	Converter	Rated Power	1725kW	
		DC Voltage Range	1500Vdc	
		AC Rated Voltage	690Vac	
		Maximum Efficiency	99%	
		Quantity	2	
	Anti-Corrosion		C4	
	IP Rating		IP55	
	Dimensions (W×H×D)		6058×2896×2438mm	
Weight		18,000Kg		
Compliance		GB17467, IEC62271-202		
Communication Interfaces		RS485/CAN/Ethernet		
Communication Protocols		Modbus-RTU/Modbus-TCP/IEC61850/IEC104		
Operating Temperature Range		-25℃~60℃		
Max. Working Altitude		4000m		
DC Side				
Container Option	Nameplate Energy		3.44MWh	
	Dimension (W×H×D)		6058×2896×2438mm	
	Communication Interfaces		RS485/CAN/Ethernet	
	Communication Protocols		Modbus-RTU/Modbus-TCP	
	Operating Temperature Range		-30℃~50℃	
	Max. Working Altitude		3000m	
	Weight		34,000Kg	
	Quantity		4	

DC Product: Liquid-Cooling Cabinet



Highlights

- Low Capex
- Easy Transportation
- Flexible Depolyment According to the Site Layout
- Low Field Labor Cost

System Features



Rack level Lockable Disconnect



Short Circuit Protection



Fire Detection and Suppression



Water-based Supression System (Optional)



Intrusion Detect System



System Controller

Technical Specification

System Model	SU344E170KM	SU344E340KM
System Information		
Nominal Power	172kW	344kW
Nameplate Energy	344kWh	
Battery Information		
Battery Chemistry	LFP	
Capacity	280Ah	
Configuration	384S1P	
Nominal Voltage	1228.8V	
Voltage Range	1075.2~1382.4V	
Working Conditions		
Degree of Protection	NEMA 3R / IP55	
Noise Emission	≤75dB@1M	
Operating Temperature Range	-30℃~50℃	
Relative Humidity	0~95% (No Condensing)	
Max. Working Altitude	3000m	
General Information		
Dimensions(W×H×D)	1300×2430×1300mm	
Weight	3,500Kg	
Cooling Method	Liquid Cooling	
Fire Safety	Aerosol / Deflagration Vent / Dry Pipe	
Certificate	UL1973, UL9540A, IEC62619, IEC61000	

DC Product: Liquid-Cooling Container



Highlights

- Superior Thermal Management
- Long Cycle Life
- Outstanding Safety Design

System Features

- 

Rack level Lockable Disconnect
- 

Short Circuit Protection
- 

Fire Detection and Suppression
- 

Water-based Supression System (Optional)
- 

Intrusion Detect System
- 

System Controller

Technical Specification

System Model	SU3441E1725KC	SU3441E3440KC
System Information		
Nominal AC Power	1725kW	3440kW
Nameplate Energy	3440.64kWh	
Battery Information		
Battery Chemistry	LFP	
Capacity	280Ah	
Configuration	384S10P	
Nominal Voltage	1228.8V	
Voltage Range	1075.2~1382.4V	
Working Conditions		
Degree of Protection	NEMA 3R / IP55	
Noise Emission	≤80dB@1M	
Operating Temperature Range	-30℃~50℃	
Relative Humidity	0~95% (No Condensing)	
Max. Working Altitude	3000m	
General Information		
Dimensions(W×H×D)	6058×2896×2438mm	
Weight	34,000Kg	
Cooling Method	Liquid Cooling	
Fire Safety	Aerosol / Deflagration Vent / Dry Pipe	
Aux.Power Supply	400VAC 50Hz 3P4L	
Aux.Current	72A	90A
Communication Protocol	Modbus TCP/IP/IEC61850/IEC104	
Certificate	UL1973, UL9540A, IEC62619, IEC61000, GB/T 36276	

DC Product: Liquid-Cooling Container



Highlights

- Highest Energy Density
- More than 40% Site Footprint Reduction
- Cutting-edge Design Enables Back-to-Back and End-to-end Layout

System Features



Rack level Lockable Disconnect



Short Circuit Protection



Fire Detection and Suppression



Water-based Supression System (Optional)



Intrusion Detect System



System Controller

Technical Specification

System Model	SU5016E2500KC
System Information	
Nominal AC Power	2500kW
Nameplate	5015.96kWh
Battery Information	
Battery Chemistry	LFP
Capacity	314Ah
Configuration	12P416S
Nominal Voltage	1331.2V
Voltage Range	1164.8~1497.6V
Working Conditions	
Degree of Protection	NEMA 3R / IP55
Noise Emission	≤ 80dB@1M
Operating Temperature Range	-30℃~50℃
Relative Humidity	0~95% (No Condensing)
Max. Working Altitude	3000m
Information	
Dimensions(W×H×D)	6058×2896×2438mm
Weight	43,000±500Kg
Cooling Method	Liquid Cooling
Fire Safety	Dry pipe / Gas detection and Ventilation / Aerosol (Optional)
Aux.Current of Aux. Power Supply	400VAC 50Hz 3P4L
Communication Protocol	Modbus TCP/IP
Certificate	UL1973, UL9540A, IEC62619, IEC61000, GB/T 36276, UL9540(Pending)



Highlights

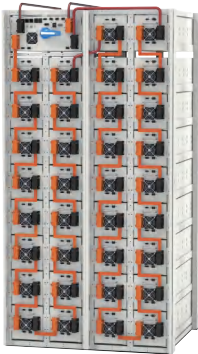
- Black start Capability
- Support multiple PCS paeallel for off-grid Operation
- Peak efficiency 99% with three-level Topology Derivation
- Strong Environment adaptability
- Excellent thermal design. No derating below 45°C ambient temperature
- Strong overload capability. supporting 1.1 times overload 10min, 1.2 times overload 1min

Technical Specification

System Model		SYL-PCS-1000K	SYL-PCS-1250K	SYL-PCS-1750K	
Operating Mode		Grid tie on / Off-Grid Mode			
DC Parameters	DC Voltage Range		580V~1500V	725V~1500V	1000V~1500V
	Max. DC Current		1935A	1935A	1935A
	Max. DC Power		1100kW	1375kW	1925kW
	Nominal Power		1000kW	1250kW	1750kW
AC Parameters	Grid tie Mode	Nominal AC Power	1000kW	1250kW	1750kW
		Max. AC Power	1100kW	1375kW	1925kW
		Nominal Voltage	400V	500V	690V
		Operating Frequency	50Hz	50Hz	50Hz
		Over-Current Capability	2163A(rms)	2163A(rms)	2163A(rms)
		Reactive Power Range	-1000kVar~1000kVar	-1250kVar~1250kVar	-1750kVar~1750kVar
		Rated Power Factor	> 0.99	> 0.99	> 0.99
		Adjustable Range of Power Factor	±1	±1	±1
		Permissible Grid Frequency Deviation	45~55Hz	45~55Hz	45~55Hz
	Permissible Grid Voltage Deviation	340V~440V	425V~550V	607V~759V	
	Access Method	3P3L	3P3L	3P3L	
	Off-Grid Mode	Rated Output Voltage	400V	500V	690V
		Rated Output Frequency	50Hz	50Hz	50Hz
		Rated Output Power	1000kW	1250kW	1750kW
		Max. Output Power	1100kW	1375kW	1925kW
		Reactive Power Range	-1000kVar~1000kVar	-1250kVar~1250kVar	-1750kVar~1750kVar
		Rated Power Factor	> 0.99	> 0.99	> 0.99
		AC Output Overload Protection Value	2163A(rms)	2163A(rms)	2163A(rms)
AC Output Wiring Method		3P3L	3P3L	3P3L	
Electrical Isolation Type		Non-Isolation			
Performance Parameter	Voltage control (Identification) Accuracy		0.1V		
	Frequency Control (Identification) Accuracy		0.001Hz		
	Idle Power Consumption		< 100W		
	Cooling		Smart-Air-Cooling		
	Dimension (W×H×D)		1475×2386×1100mm		
	Weight		1450kg		
Working Conditions	Operating Temperature Range		-40℃~+60℃		
	Storage Temperature Range		-40℃~+70℃		
	Electromagnetic environment category		A		
	Relative Humidity		0~95%		
	Degree of Protection		IP54		
	Max. Working Altitude		4000m		
Communication	External communication interface		RS485, CAN, Ethernet		
	External communication protocol		MODBUS-TCP		
Response time		100ms			
Certificate		IEC62477, GB/T 34120-2017, GB/T 34133-2017, CGC-R46103:2018A			

280Ah Energy Storage Battery System Specification

Cell	Specification
Battery Chemistry	LFP
Cell Type	Prismatic
Dimension (W×H×D)	174.7×71.65×207.11mm
Weight	5.43kg
Nominal Capacity	280Ah
Nominal Energy	896Wh
Nominal Voltage	3.2Vdc
Operating Voltage Range	2.8~3.6Vdc
Pack	
Configuration	1P10S
Key Components	10Cells,Module BMU
Dimension (W×H×D)	220×232×980mm
Weight	72kg
Nominal Capacity	280Ah
Nominal Energy	8.96kWh
Nominal Voltage	32V
Operating Voltage Range	28~36Vdc
Maximum Power	4.48kW (0.5C)
Rack	
Configuration	1P340S
Key Components	34Modules, 1Bspu
Dimension (W×H×D)	1,160×2,275×1,000mm
Weight	2,800kg
Nominal Capacity	280Ah
Nominal Energy	304.64kWh
Nominal Voltage	1,088V
Operating Voltage Range	952~1,224V
Maximum Power	152.32kW(0.5C)
Certificates	IEC62619, IEC61000, UL1973,UL9540A, UN38.3



280Ah Energy Storage Battery System Specification

Cell	Specification
Battery Chemistry	LFP
Cell Type	Prismatic
Dimension (W×H×D)	174.7×71.65×207.11mm
Weight	5.43kg
Nominal Capacity	280Ah
Nominal Energy	896Wh
Nominal Voltage	3.2Vdc
Operating Voltage Range	2.8~3.6Vdc
Pack	
Configuration	1P48S
Key Components	48Cells,Module BMU
Dimension (W×H×D)	810×254.5×1111.5mm
Weight	310kg
Nominal Capacity	280Ah
Nominal Energy	43.008kWh
Nominal Voltage	153.6V
Operating Voltage Range	134.4~172.8
Maximum Power	43.008kWh(1C)
Rack	
Configuration	1P384S
Key Components	8Modules, 1Bspu
Dimension (W×H×D)	950×2480×1120mm
Weight	3100kg
Nominal Capacity	280Ah
Nominal Energy	344.064kWh
Nominal Voltage	1228.8V
Operating Voltage Range	1075.2~1382.4V
Maximum Power	344.064kW(1C)
Certificates	UL1973, UL9540A, GB/T36276 IEC62619, IEC61000, IEC62477, UN38.3



Striving to be the world's top BESS solution provider of all applications

Global Sales Network

- ▲ Headquarters
- Sales Service Center



Global Certification



Case Reference



100MW/123MWh

Address: USA

Time: 2023

Application:

Frequency Regulation



25MW/117.5MWh

Address: USA

Time: 2022

Application:

Energy Shifting
Renewable Smoothing



25MW/116MWh

Address: Belgium

Time: 2023

Application:

Energy Shifting
Renewable Smoothing

40MW/109MWh

Address: USA

Time: 2022

Application:

Energy Shifting
Renewable Smoothing



22.7MW/22.7MWh

Address: UK
Time: 2023
Application: Frequency Regulation



200MW/400MWh

Address: China
Time: 2022
Application: Grid-Side Shared Storage
Renewable Smoothing



200kW/502kWh

Address: Germany
Time: 2023
Application: C&I



50MW/245MWh

Address: USA
Time: 2023
Application:
Energy Shifting
Renewable Smoothing



Case Reference



Time: 2022, Address: USA, Size: 5.48MWh



Time: 2021, Address: Australia, Size: 8.6MWh



Time: 2022, Address: China, Size: 14MW/14.32MWh



Time: 2021, Address: USA, Size: 47MWh



Time: 2021, Address: China, Size: 7.5MWh



Time: 2021, Address: China, Size: 5MW/10MWh



Time: 2020, Address: USA, Size: 6.4MWh



Time: 2023, Address: UK, Size: 57MW/55.91MWh



Time: 2021, Address: Poland, Size: 15.5MWh